US Department of Agriculture (USDA)



Statement of Work

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Airport BAR- Seed Collection

General Information

1.0 Scope of Work

The intent of this project is to procure services to locate, identify and map collection sites of native plant species; to collect seed from these native species; to clean, store, and deliver the collected seed to the Cleveland National Forest to support post-fire restoration projects within the Airport Fire burn area. Seed collection will occur on the Trabuco Ranger District of the Cleveland National Forest. The contractor will work and communicate with Forest staff to determine and identify collection species and sites. **Up to approximately 100-300 lbs of native plant seed** shall be collected during this project.

The Contractor will be responsible for the following:

1. Collect seeds of native plant species, following seed collection standards in Section C.6-2 Seed Collection Standards.
2. Clean seed and test seed viability, purity, and germination of the products, following standards in Sections C.6-3 Seed Testing Standards and C.6-4 Seed Cleaning and Storage Standards. Seed viability test reports will be distributed to the Forest and COR.
3. Package and store seed in appropriate conditions outlined in Section C.6-4 Seed Cleaning and Storage Standards, until distributed to the Forest. Standards for shipping native plant materials and seed outlined in Section C.6-5 Seed Delivery Standards.

Access to collection areas is by vehicle and foot. Vehicle access includes a variety of road conditions including ungraded dirt roads. Roads are frequently traveled and used by forest staff and neighboring residents, so care will be needed in these areas. Work may occur on rough, uneven and remote terrain and will require hiking.

The Cleveland National Forest will be ordering a seed collection crew to collect native plant seeds in the areas shown in Fig 1. The days worked may be non-contiguous.

The contractor will provide:

1) A survey crew of at least two (2) qualified botanists with the ability to identify target plant species, including non-native, invasive weeds. Work may occur in remote locations requiring driving on rough, infrequently maintained dirt roads and hiking on uneven terrain, riparian corridors, and steep slopes.
2) A seed collection crew of 2 to 6 seed collection technicians to collect seeds from native plants during the appropriate season within the project area. This crew will consist of at least one experienced botanist/specialist who will be able to identify target species for collection. Some work may occur in remote locations requiring substantial driving on rough, infrequently maintained dirt roads and hiking on uneven terrain, riparian corridors, and steep slopes.

2.0 Background

***Location*:** The project is located on the Trabuco Ranger District of the Cleveland National Forest, in and around the Airport Fire footprint (Fig. 1).

Accessibility: Most work areas are accessed by the road system and by hiking. Keys may be provided by the local forest to access additional areas.

Description: The 2024 Airport Fire burned approximately 23K acres on the Trabuco Ranger District of the Cleveland National Forest. The fire mostly burned grassland and chaparral areas. Meadow and grassland areas were infested with non-native grasses and forbs after the fire, and there is little persistent native vegetation regeneration in these areas. There is a need for additional support in native vegetation regeneration, and native plant seed collection in and around the fire footprint will help provide native plant seed to sow in areas that are struggling with natural native vegetation regeneration.

Approximately **100-300 lbs of native plant seed is needed to support native vegetation regeneration in post-fire meadow and grassland areas.** Target native plant species listed in Table 1.

Maps: A general map of the work area is attached as Figure 1. Additional GIS shapefiles and layers will also be provided during the pre-work meeting.

Contractor Requirements

3.0 Technical Requirements / Tasks

3.1. General Requirements

- The Contractor shall furnish the necessary personnel, supplies, materials, equipment, transportation, services and facilities to perform these specifications.
- The Contractor shall communicate with the COR throughout the duration of the project.
- All vehicles must stay on designated FS roads. No off-road vehicle driving is authorized for seed collection.
- The Contractor shall *avoid* all flagged locations of heritage resources and if any heritage resources are discovered during implementation, all project related work shall *cease* immediately and notification to COR initiated.

3.2. Seed Collection Standards

The Contractor shall perform the following tasks as part of seed collection:

A. **Seed Reconnaissance**: Conduct reconnaissance trips to find and evaluate collection populations and seed maturity. Maps may be provided by the local forest; however, the Contractor is expected to locate and identify additional seed collection sites to meet seed collection needs.
B. **Species Identification**: All species identification will be confirmed using an approved source (i.e. Flora of North America, The Jepson Manual Vascular Plants of California). Collectors shall be able to identify target collection species and non-target plants to species level. The Contractor shall provide FS with pictures of each species being collected or collect a voucher herbarium specimen (as determined as species are selected) that show enough characteristics to identify the plant and ensure the correct species is being collected.
C. **Stage of Maturity**: All species shall be harvested when the seed maturity is at the hard dough stage or later. The stage of seed maturity should be checked by cross sectioning several seeds from each site using a knife or fingernail clippers and a hand lens to confirm maturity. If seed is at an early hard dough stage contractor will collect a larger portion of the stem and allow seed to dry and mature while attached to the stem.
D. **Seed collection** in the field, within appropriate seed zones and on FS lands only. Exact timing of collection trips, species, and locations will vary and be dependent on seed production and maturity. Target collection species and areas for collection shall be determined upon award of the agreement by the FS. Specific seed collection standards will be followed:
    1. Mature, sound seed will be collected by hand. Seed is considered sound when the bud endosperm and embryo are developing normally and there is no evidence of insect, disease, climatic or other types of damage.
    2. Percentage of seed taken from an individual shall be determined in consultation with FS (typically no more than 25%).
    3. No ground disturbance will be incurred. No disturbance to plants from which seeds are collected will be incurred. Significant damages (such as cut limbs or crushing) to the parent plant should not be incurred during seed collection.
    4. Seed heads may be cut off just below the flower inflorescence, however cutting shears must be sterilized between collection areas to prevent disease spread.
    5. Care must be taken to avoid weedy areas in order to avoid seed contamination and the spread of non-natives.

6. No off-road vehicle driving is allowed for collection. Vehicles must stay on designated roadbeds that are not overgrown with vegetation.
7. For each species, seed shall be obtained from more than three populations within the permitted collection area. A different population is defined here as two populations that have a very low chance of cross-pollinating with each other. In other words, they are separated by distance or some type of geographical barrier. In order to make this determination it is often necessary to have some knowledge of the species' main pollinator and its movement capabilities.
8. Seed shall be collected from at least 50 different, well dispersed individuals in a population.
9. Do not collect seed in the rain, snow, or periods of heavy dew.
10. Keep each bag of collected seed separate. Do not bulk seed together into larger bags. This is done as a precaution to isolate seed collections in case a seed lot is found to be contaminated with other species and help in maintaining seed quality.
11. Seeds will be placed in paper (or industry standard cloth or seed bags) bags with sufficient air exchange and labeled with:
    - Species name
    - Collection site= Forest of origin and GPS coordinates of collection location
    - Pure live seed (PLS) target species weight (pounds or ounces)
    - Non-target species weight (ounces)
    - Plant material debris weight (ounces)
    - Date collected and collector's name
    - Any applied pre-treatments for germination
    - The bag will be secured to reduce material loss. See Section C.6-4 for seed cleaning and storage standards.

E. Collection locations (both general location descriptions and coordinates/UTM's) will be recorded for all species collected. GPS location of each collection site. Label each bag with the species and collection location. Keep seed from each collection site separate. Data will be submitted in a GIS format. Shapefile or geodatabase is acceptable.

F. Purity: Take precautions to ensure that non-target collection species are not accidentally collected and mixed during the collection process. Often it is impossible to separate similar species once mixed together.

G. Weed Prevention: Ensure all vehicles, gear, tools, equipment, personnel (e.g., clothing, boots, laces), PPE, etc. involved with or otherwise used in project implementation arrive to the project site clean and free of off-site soil and/or organic debris.

H. The Contractor shall collect seeds from potential native plant species listed below in Figures and Tables- see Table 1. **Species list may be flexible and subject to change throughout the duration of the project due to species and seed availability. Discussion with local forest botanist/s and COR is required prior to making any changes in the species collection list.**

3.3. Seed Testing Standards/Requirements

A. All seeds collected from forest sites and seeds harvested will be cleaned and tested for viability, purity, and germination by either University of California Davis Seed Biotechnology Center, or a California state accredited facility, approved by the respective forest botanist.

B. All results from seed viability, purity, and germination tests will be sent to the respective forest POCs (Table 1) and the COR.

3.4. Seed Cleaning and Storage Standards

A. **All seed collected from the field** shall be cleaned of chaff and excess plant material. No more than 15% of the final cleaned seed weight shall be comprised of this material.

B. **All seed collected from the field** shall be cleaned of non-target species. No more than 5% of the final cleaned seed weight shall be comprised of non-target species.

C. The Contractor shall fill out two, identical seed identification tags and/or labels for each collected seed bag/container. The Contractor shall place one tag inside the seed package and the other tag shall be securely attached to the outside of the package. The outside of each package of seed shall also be marked with the species name and collection site information. Seed identification tags and/or labels will include:

1. Species name
2. Collection site= Forest of origin and GPS coordinates of collection location
3. Pure live seed (PLS) target species weight (pounds or ounces)
4. Non-target species weight (ounces)
5. Plant material debris weight (ounces)
6. Date collected and collector's name
7. Any applied pre-treatments for germination

D. Seed bags will be kept in a cool, dry place during all phases of collection, handling, cleaning and storage. Do not freeze seed, and consult respective forest prior to freezing any seed, if that is recommended.

E. Storage of plant materials collected from the field: To prevent damage of seed by mold development, at the end of each collection day, air dry collected seed. Do not store freshly collected seed in vehicles in full sun. Properly store all plant materials (bulk and cleaned seed). Bulk material shall be stored such that it has good ventilation and dries properly per standard plant material and seed handling procedures. Take precautions to prevent seed damage from insects by considering the use of bug strips or other effective methods.

F. **All seed collected from the field** by the Contractor shall be protected from all forms of potential damage such as: moisture, heat, chemicals, insects, mechanical and other environmental conditions detrimental to healthy, viable seed.

3.5. Seed Delivery Standards

The Contractor will have experience and availability of increase fields throughout the duration of the contract. The Contractor is requested to perform the following tasks: cleaning, testing, and storing native seed; preparing seed accessions and delivery of harvested seed:

A. **Seed Storage:** Seed shall not be allowed to become wet after harvest. Seed shall be stored in a building, shed or warehouse that allows air circulation and is protected from rain, moisture, temperature extremes and rodents, insect and pathogen attack, or other conditions that would reduce the viability of the seed.

B. **Target Seed Purity Standards:** The following general target standards shall apply to all seed collected. Note these are standards to strive to achieve.
   i. Weed Seeds maximum 0.1%
   ii. Noxious weed seed 0%
   iii. No “off type” *Bromus* species seeds

The Contractor shall provide, in writing, any reasons for seed not meeting these targets. If fields are enrolled in certification through the Pre-Varietal Germplasm Program, standards are set according to species, and in most cases, are more rigid than the general standards listed above.

C. **Seed Shipment:** Seed shall be shipped in a manner that will prevent damage to the seed and arrive expeditiously at the forest. The Contractor shall provide notification to FS before the seed is shipped and shall provide the tracking number of the package by email as soon as is practical. The respective forest botanist shall provide shipping addresses for forests receiving seeds.

D. **Seed Delivery:** All seed delivered shall conform to all laws and regulations pertaining to the shipment of seed required by the Federal Seed Act and the Seed laws pertaining to the representative state. Submit original seed test analysis and copy of the seed tags by lot number to FS. Seed shall be shipped in a manner to prevent damage to the seed.

4.0 Government Furnished

- The Government will provide gate key for vehicular access to some work sites.
- The Government will also provide GIS data of infestations, project areas and land ownership.

5.0 Deliverables & Timeline

Estimated Start Date & Contract Time

Projected Start Date: **November 30th, 2026** *Projected Completion Date:* **September 30, 2027**

Work will occur during optimal times to collect seeds from target priority species (see Table 1).

Project Timeline

Year	Task/Activity	Timing
1	Seed collection reconnaissance	November 2026- June 2027
1	Seed collection	May 2027- September 2027
1	Seed cleaning	May 2027- September 2027
1	Seed testing and packaging	May 2027- September 2027
1	Seed delivery and reporting	May 2027- September 2027

Deliverables:

A. Clean seed, appropriately labeled and packaged, as indicated in Sections 3.3 and 3.4.

B. All seed viability, germination, and purity test reports for forest collected seeds, as indicated in Sections 3.3 and 3.4.

C. Any leftover collected seed, cleaned, packaged, and labeled, as indicated in C.6-4.

D. Brief summary of work/project activities completed, may include:

   a. Description of major findings
   b. List of species collected, total amounts collected, and collection locations
   c. Seed viability results

E. Timeline for deliverables (dates are flexible):

Deliverable	Due date
Seed viability, germination, and purity test results for forest collected seed	September 1, 2027
Cleaned seed, delivered to forest	September 30, 2027

6.0 Government Inspection & Acceptance

A. Collection records will be reviewed for documentation of activities and time spent in reconnaissance/survey work and seed collection work. COR, respective forest botanist/s, and/or technical monitors will make field site visits throughout the project.
B. Seed packages will be visually inspected for debris and chaff content; seed species; signs of disease; and insect and mechanical damage.
C. Seed viability, germination, and purity test results will be submitted to forest botanists and COR prior to seed delivery, to determine seed acceptance.
D. **Unacceptable Seed** - The Government will not accept seed under the following conditions:
    1. Seed that is not properly packaged and identified according to the specifications in Section C.6-5 of this contract.
    2. Seed packages that contain more than a total of 5 percent by weight of seed from any other non-target plant species.

6.1. Re-Work

Unacceptable seed will be rejected by the Government. The Contractor shall be responsible for either re-cleaning or re-collecting seed to meet the contract specifications. All re-work shall be at no additional cost to the Government.

6.2. Payment

Payment will be made for work performed as follows: location, field documentation, mapping, seed collection, seed cleaning, seed packaging, collection summary, and travel related to, will be paid.

7.0 Performance Standards

Contractor performance/s shall be described/evaluated by the following:

Excellent

Performance meets contractual requirements and exceeds many requirements that benefited the Forest. Work was accomplished with few, if any, minor problems for which corrective actions taken by the contractor were highly effective.

Good

Performance meets contractual requirements and exceeds some requirements that benefit the Forest. Work was accomplished with few problems for which corrective actions taken by the contractor were effective.

Acceptable

Performance meets contractual requirements and work was accomplished with minor problems for which corrective actions were taken by the contractor were satisfactory to the Forest.

Marginal

Performance does not meet some contractual requirements and serious problems with contractor performance were documented. Corrective actions were not taken or appear to be marginally effective.

Unacceptable

Performance did not meet most contractual requirements and serious issues with the performance were experienced, for which corrective actions were not taken and/or were ineffective.

Performance Requirements Summary (PRS) Table

Performance Objective	Performance Standard	Acceptable Quality Level (AQL)	Method of Surveillance	Remedy for Non-Performance

1. Seed collection	- collect seeds from approved areas	Collect 50% or more of species on the species list.	COR or Forest staff may conduct field visits and check-ins.	COR or Forest staff may ask for follow up information/details.
2. Seed cleaning and viability testing	- clean seed - submit viability testing results by Oct. 1st 2027/ in a timely manner.	Timely delivery/submittal of viability testing report to COR or Forest staff	COR or Forest staff will review and approve reports.	COR or Forest staff may ask for follow up information/details.
3. Seed delivery and reporting	- deliver seed to forest by Oct. 31st 2027 - submit invoice/s	Timely delivery of seed and invoices	email communication to COR and CO	Discussion with COR, CO, and contractor to fix situation.

Figures and Tables

Table 1. Native plant species recommended for seed collection. Discussion with local forest botanist/s is needed for additional clarification on exact species and approximate amounts (lbs) to be collected.

Lifeform	Species	Common Name	LBS
grass	*Bromus stichens*	sitka brome	10
grass	*Carex praegracilis*	sedge species	3
grass	*Eleocharis macrostachya*	spike rush	2
grass	*Elymus glaucus*	blue wildrye	5
grass	*Juncus balticus*	wire rush	5
grass	*Juncus textilis*	basket rush	5
grass	*Koeleria macrantha*	june grass	5
grass	*Melica imperfecta*	California melic	5
grass	*Muhlenbergia rigens*	deergrass	5
grass	*Stipa coronata*	crested needle grass	10
grass	*Stipa lepida*	foothill needle grass	10
grass	*Stipa pulchra*	purple needle grass	10
herb	*Collinsia heterophylla*	Chinese houses	1
herb	*Acourtia microcephala*	sacapellote	2
herb	*Agoseris grandiflora*	giant mountain dandelion	2
herb	*Artemisia douglasiana*	mugwort	3
herb	*Bloomeria crocea*	golden stars	1
herb	*Calandrinia menziesii*	red maids	1
herb	*Calystegia macrostegia*	morning glory	5
herb	*Castilleja exserta*	owl's clover	1
herb	*Chaenactis artemisiifolia*	Artemisia leaved chaenactis	3
herb	*Cirsium occidentale*	western thistle	2
herb	*Clarkia purpurea*	wine cup clarkia	5
herb	*Claytonia perfoliata*	miner's lettuce	2
herb	*Collinsia concolor*	southern Chinese houses	1
herb	*Corethrogyne filaginifolia*	sand aster	8
herb	*Croton setiger*	doveweed	8
herb	*Cryptantha sp*	popcorn flower species	5
herb	*Deinandra kelloggii*	Kellogg's tarweed	2

herb	*Dipterostemon capitatus ssp. capitatus*	wild hyacinth	2
herb	*Drymocallis glandulosa var. glandulosa*	sticky cinquefoil	2
herb	*Epilobium canum*	California fuchsia	3
herb	*Erhendorferia chrysantha*	golden eardrops	5
herb	*Erigeron foliosus var. foliosus*	threadstemmed fleabane	5
herb	*Eriogonum elongatum var. elongatum*	long stemmed buckwheat	3
herb	*Eriogonum gracile*	slender buckwheat	3
herb	*Gutierrezia sarothrae*	matchweed	5
herb	*Helianthus gracilentus*	slender sunflower	10
herb	*Heterotheca grandiflora*	telegraph weed	10
herb	*Heterotheca sessiliflora*	golden aster	3
herb	*Lasthenia gracilis*	needle goldfields	1
herb	*Lupinus bicolor*	miniature lupine	2
herb	*Lupinus concinnus*	Bajada lupine	1
herb	*Lupinus excubitus*	grape lupine	5
herb	*Madia gracilis*	gumweed	1
herb	Malacothrix saxatilis	cliff aster	2
herb	*Nemophila menziesii*	baby blue eyes	1
herb	*Phacelia imbricata*	imbricate phacelia	5
herb	*Phacelia minor*	blue bells	2
herb	*Salvia columbariae*	chia	3
herb	*Stephanomeria diegensis*	San Diego wreathplant	4
herb	*Trichostema lanceolatum*	vinegarweed	2
herb	*Wyethia ovata*	mules ears	2
shrub	*Artemisia californica*	California sagebrush	2
shrub	*Dendromecon rigida*	bush poppy	8
shrub	*Eriogonum fasciculatum*	California buckwheat	20
shrub	*Frangula californica*	coffeeberry	10
shrub	*Lonicera subspicata*	southern honeysuckle	4
shrub	*Lupinus albifrons var. austromontanus*	southern mountain lupine	5
shrub	*Lupinus hirsutissimus*	stinging lupine	5
shrub	*Lupinus truncatus*	blunt leaved lupine	3
shrub	*Paeonia californica*	California peony	4
shrub	*Quercus berberidifolia*	scrub oak	10
tree	*Quercus agrifolia*	coast live oak	20

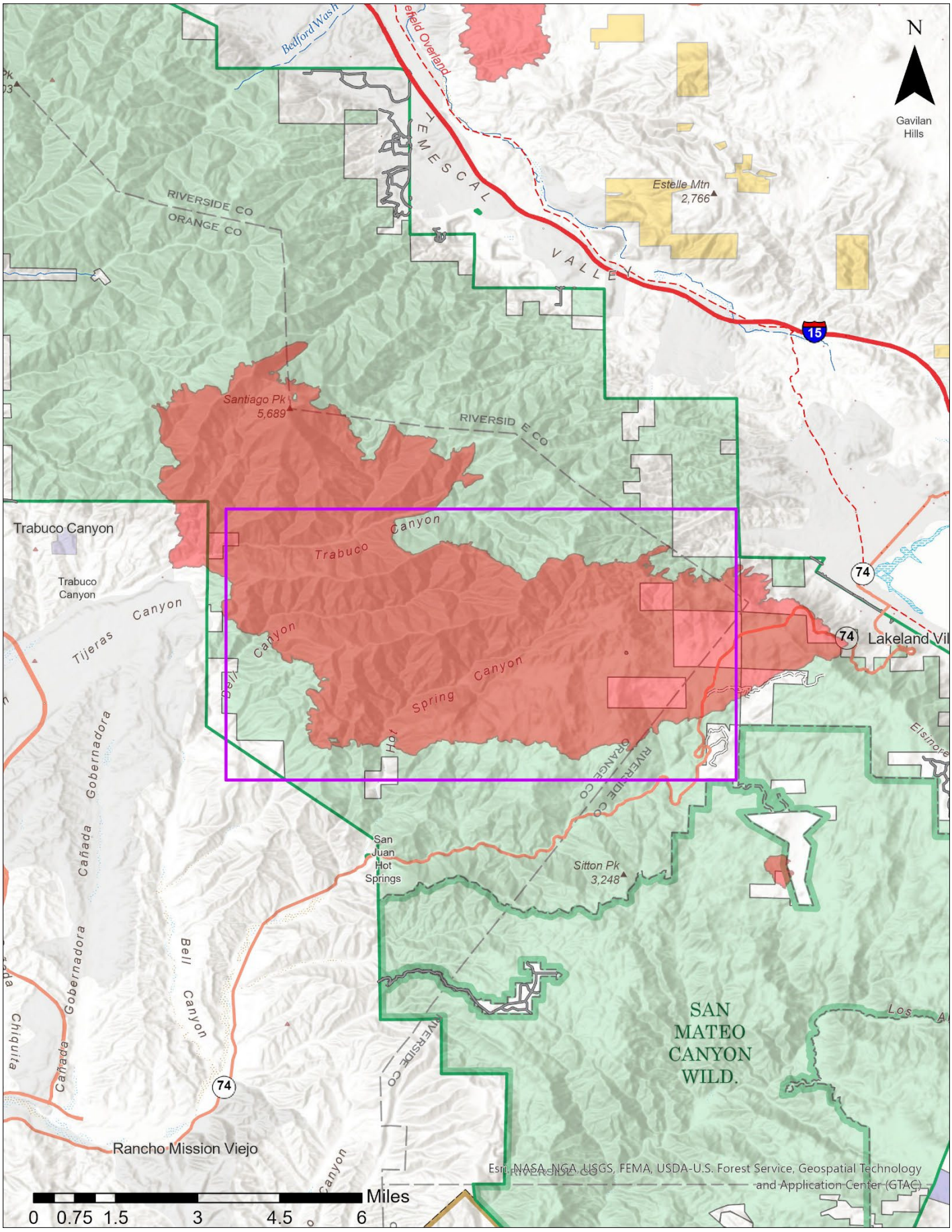


Fig. 1. Map of Seed Collection Area. Seed collection area is outlined in purple. Seed collection may only occur on Forest Service lands